

Ex.no:5B	DATA VISUALIZATION USING MATPLOTLIB

Aim:

To visualize the Daily Google Search Trends US dataset using line charts, bar charts, pie charts, histograms, scatter plots, and box plots with Matplotlib.

Algorithm:

ALGORITHM 1: LINE CHART

Objective: To visualize the number of Google Search Trends across different years.

Steps

1. Load the Daily Google Search Trends US dataset.
2. Group the data by Year and count the number of search trends.
3. Arrange the yearly trend counts in chronological order.
4. Plot the number of search trends using a line chart.
5. Add title, axis labels, and grid.
6. Display the graph.

Interpretation

The line chart shows how the number of Google Search Trends varies across different years and helps identify overall trends.

```
year_trends = df.groupby("year") ["query"].count().sort_index()
plt.figure(figsize=(6,4))
plt.plot(year_trends.index, year_trends.values, marker='o')
plt.title("Year vs Number of Search Trends")
plt.xlabel("Year")
plt.ylabel("Number of Search Trends")
plt.grid(True)
plt.show()
```

ALGORITHM 2: BAR CHART

Objective: To compare the number of Google Search Trends across different categories.

Steps

1. Select the Category column.
2. Group the data by Category and count the number of search trends.
3. Create a bar chart using the category names and their trend counts.
4. Add title and axis labels.
5. Display the chart.

Interpretation

The bar chart provides a comparison of the number of Google Search Trends among different categories in the dataset.

```
category_counts = df["category"].value_counts()
plt.figure(figsize=(8,4))
plt.bar(category_counts.index, category_counts.values)
plt.title("Google Search Trends by Category")
plt.xlabel("Category")
plt.ylabel("Number of Trends")
plt.xticks(rotation=45)
plt.show()
```

ALGORITHM 3: PIE CHART

Objective: To visualize the distribution of Google Search Trends across categories.

Steps

1. Select the Category column.
2. Count the occurrences of each category.
3. Calculate their proportions.
4. Create a pie chart.
5. Display percentage labels.
6. Add a suitable title.

Interpretation

The pie chart shows the percentage contribution of each category to the total Google Search Trends.

```
category_counts = df["category"].value_counts()
plt.figure(figsize=(6,6))
plt.pie(category_counts, labels=category_counts.index, autopct="%1.1f%%",
startangle=90)
plt.title("Google Search Trends Category Distribution")
plt.show()
```

ALGORITHM 4: HISTOGRAM

Objective: To visualize the distribution of search query ranks.

Steps

1. Select the Rank column.
2. Remove missing Rank values.
3. Divide the Rank values into intervals or bins.
4. Create a histogram using Matplotlib.
5. Add title and axis labels.
6. Display the histogram.

Interpretation

The histogram shows the frequency distribution of search query ranks and helps understand their spread.

```
plt.figure(figsize=(6,4))  
plt.hist(df["rank"].dropna(), bins=10)  
plt.title("Search Query Rank Distribution")  
plt.xlabel("Rank")  
plt.ylabel("Frequency")  
plt.show()
```

ALGORITHM 5: SCATTER PLOT

Objective: To analyze the relationship between Year and Search Query Rank.

Steps

1. Select Year as the X-axis variable.
2. Select Rank as the Y-axis variable.
3. Plot the observations using a scatter plot.
4. Add title and axis labels.
5. Display the graph.

Interpretation

The scatter plot helps identify the visible relationship between Year and Search Query Rank in the dataset.

```
plt.figure(figsize=(6,4))  
plt.scatter(df["year"], df["rank"])  
plt.title("Year vs Search Query Rank")  
plt.xlabel("Year")  
plt.ylabel("Rank")  
plt.show()
```

ALGORITHM 6: BOX PLOT

Objective: To visualize the distribution and identify possible outliers in Search Query Rank.

Steps

1. Select the Rank column.
2. Remove missing values if present.
3. Create a box plot using Matplotlib.
4. Add a suitable title and axis label.

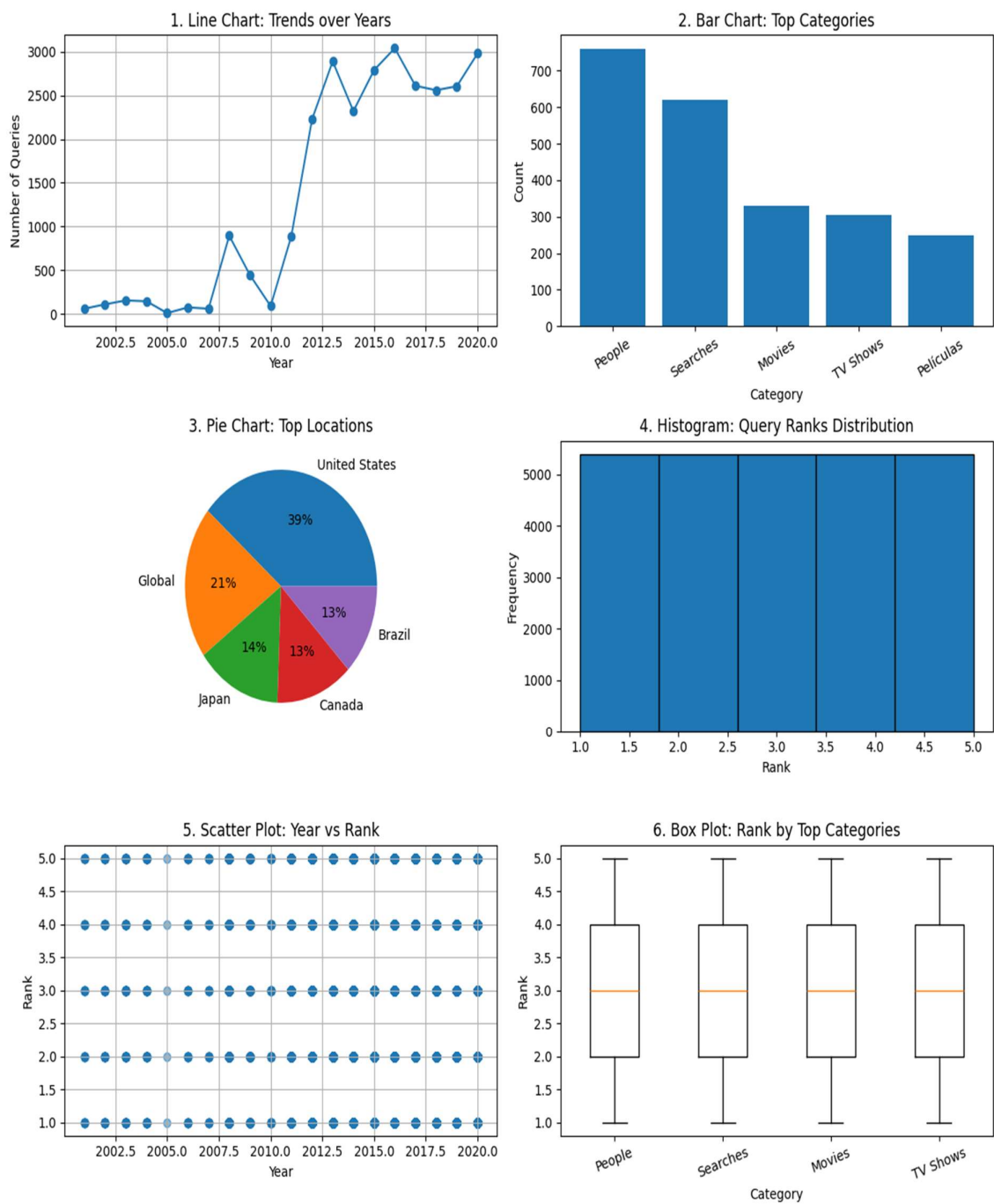
5. Display the plot.

Interpretation

The box plot shows the central distribution of Search Query Rank and helps identify unusually high or low rank values.

```
plt.figure(figsize=(4,5))  
plt.boxplot(df["rank"].dropna())  
plt.title("Search Query Rank Box Plot")  
plt.ylabel("Rank")  
plt.show()
```

Output:



Result: